

Machine Learning Predictions of Oil Yields Obtained by Plastic Pyrolysis and Application to Thermodynamic Analysis

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Section SI.1 Data

The data set used for the modeling in this work can be seen below in Table SI.1. There are 325 data points pulled from 39 papers published in the open literature.

Table SI.1 Data set of 325 data points used in modeling

HDPE (wt%)	LDPE (wt%)	PP (wt%)	PS (wt%)	PVC (wt%)	PET (wt%)	Temperature (°C)	Heating Rate (°C/min)	Particle Size (mm)	Feed Size (g)	Catalyst	Reactor Type	Oil Yield (wt%)	Source
0	100	0	0	0	0	500	10	2.5	-	No	Fixed Bed	95	1
0	100	0	0	0	0	400	10	2.5	-	Yes	Fixed Bed	84	1
0	100	0	0	0	0	450	10	2.5	-	Yes	Fixed Bed	81	1
0	100	0	0	0	0	500	10	2.5	-	Yes	Fixed Bed	80	1
0	100	0	0	0	0	550	10	2.5	-	Yes	Fixed Bed	76	1
0	100	0	0	0	0	600	10	2.5	-	Yes	Fixed Bed	71	1
0	100	0	0	0	0	400	10	2.5	-	Yes	Fixed Bed	83	1
0	100	0	0	0	0	450	10	2.5	-	Yes	Fixed Bed	80	1
0	100	0	0	0	0	500	10	2.5	-	Yes	Fixed Bed	77	1
0	100	0	0	0	0	550	10	2.5	-	Yes	Fixed Bed	72	1
0	100	0	0	0	0	600	10	2.5	-	Yes	Fixed Bed	70	1
0	100	0	0	0	0	500	-	-	-	No	Fluidized Bed	81.2	2
0	100	0	0	0	0	550	-	-	-	No	Fluidized Bed	73.9	2
0	100	0	0	0	0	550	-	-	-	No	Fluidized Bed	72.7	2
0	100	0	0	0	0	550	-	-	-	No	Fluidized Bed	65.3	2
0	100	0	0	0	0	550	-	-	-	No	Fluidized Bed	64.7	2
0	100	0	0	0	0	600	-	-	-	No	Fluidized Bed	28.5	2
100	0	0	0	0	0	500	5	3	100	No	Fixed Bed	62	3
100	0	0	0	0	0	550	5	3	100	No	Fixed Bed	71	3
100	0	0	0	0	0	600	5	3	100	No	Fixed Bed	69	3

HDPE (wt%)	LDPE (wt%)	PP (wt%)	PS (wt%)	PVC (wt%)	PET (wt%)	Temperature (°C)	Heating Rate (°C/min)	Particle Size (mm)	Feed Size (g)	Catalyst	Reactor Type	Oil Yield (wt%)	Source
100	0	0	0	0	0	650	5	3	100	No	Fixed Bed	68	3
100	0	0	0	0	0	700	5	3	100	No	Fixed Bed	69	3
100	0	0	0	0	0	750	5	3	100	No	Fixed Bed	63	3
100	0	0	0	0	0	800	5	3	100	No	Fixed Bed	64	3
0	100	0	0	0	0	400	5	0.5	0.6	No	Batch	93.1	4
100	0	0	0	0	0	400	5	0.5	0.6	No	Batch	84.7	4
0	100	0	0	0	0	550	5	0.5	0.6	Yes	Batch	18.3	4
100	0	0	0	0	0	550	5	0.5	0.6	Yes	Batch	17.3	4
0	100	0	0	0	0	550	5	0.5	0.6	Yes	Batch	61.6	4
100	0	0	0	0	0	550	5	0.5	0.6	Yes	Batch	41	4
0	68	16	16	0	0	430	-	3	-	No	Batch	93	5
0	16	68	16	0	0	430	-	3	-	No	Batch	90	5
0	16	16	68	0	0	430	-	3	-	No	Batch	92	5
0	100	0	0	0	0	430	-	3	-	No	Batch	90	5
0	0	100	0	0	0	430	-	3	-	No	Batch	92	5
0	0	0	100	0	0	430	-	3	-	No	Batch	95	5
0	33	33	33	0	0	430	-	3	-	No	Batch	91	5
100	0	0	0	0	0	450	6.5	-	-	No	Batch	84	6
100	0	0	0	0	0	400	20	-	20	No	Batch	11.2	7
100	0	0	0	0	0	450	20	-	20	No	Batch	23.96	7
100	0	0	0	0	0	500	20	-	20	No	Batch	72.25	7
100	0	0	0	0	0	550	20	-	20	No	Batch	79.08	7
0	0	0	100	0	0	450	5	1.5	10	No	Batch	95.77	8
0	0	0	100	0	0	450	10	1.5	10	No	Batch	95.79	8
0	0	0	100	0	0	450	15	1.5	10	No	Batch	92.75	8
0	0	0	100	0	0	450	20	1.5	10	No	Batch	92.65	8

HDPE (wt%)	LDPE (wt%)	PP (wt%)	PS (wt%)	PVC (wt%)	PET (wt%)	Temperature (°C)	Heating Rate (°C/min)	Particle Size (mm)	Feed Size (g)	Catalyst	Reactor Type	Oil Yield (wt%)	Source
0	100	0	0	0	0	450	5	1.5	10	No	Batch	81.65	8
0	100	0	0	0	0	450	10	1.5	10	No	Batch	81.33	8
0	100	0	0	0	0	450	15	1.5	10	No	Batch	72.63	8
0	100	0	0	0	0	450	20	1.5	10	No	Batch	61.24	8
0	0	0	0	0	100	450	5	1.5	10	No	Batch	39.02	8
0	0	0	0	0	100	450	10	1.5	10	No	Batch	35.4	8
0	0	0	0	0	100	450	15	1.5	10	No	Batch	29.71	8
0	0	0	0	0	100	450	20	1.5	10	No	Batch	29.16	8
0	0	100	0	0	0	450	5	1.5	10	No	Batch	83.34	8
0	0	100	0	0	0	450	10	1.5	10	No	Batch	82.67	8
0	0	100	0	0	0	450	15	1.5	10	No	Batch	82.92	8
0	0	100	0	0	0	450	20	1.5	10	No	Batch	68.06	8
26.2	31.1	8.2	13	0	0	500	20	2.5	400	No	Fixed Bed	29	9
17.8	19.6	13.9	8.7	0	0	500	20	2.5	400	No	Fixed Bed	26.33	9
3.5	25	22.2	4	0	0	500	20	2.5	400	No	Fixed Bed	44.62	9
17	34	15.4	12.4	0	0	500	20	2.5	400	No	Fixed Bed	43.2	9
100	0	0	0	0	0	500	-	3	50	No	Batch	81	10
100	0	0	0	0	0	600	-	3	50	No	Batch	79.1	10
100	0	0	0	0	0	700	25	-	-	No	Fixed Bed	79.72	11
0	100	0	0	0	0	700	25	-	-	No	Fixed Bed	84.25	11
0	0	100	0	0	0	700	25	-	-	No	Fixed Bed	84.44	11
0	0	0	100	0	0	700	25	-	-	No	Fixed Bed	83.77	11
0	0	0	0	100	0	700	25	-	-	No	Fixed Bed	31.69	11
0	0	0	0	0	100	700	25	-	-	No	Fixed Bed	41.3	11
31.25	31.25	7.29	13.5	11.46	5.21	700	25	-	-	No	Fixed Bed	75.12	11
43	43	9.7	0	0	0	500	35	3.5	40	No	Horizontal Tube	63.91	12

HDPE (wt%)	LDPE (wt%)	PP (wt%)	PS (wt%)	PVC (wt%)	PET (wt%)	Temperature (°C)	Heating Rate (°C/min)	Particle Size (mm)	Feed Size (g)	Catalyst	Reactor Type	Oil Yield (wt%)	Source
43	43	9.7	0	0	0	550	50	3.5	40	No	Horizontal Tube	56.08	12
43	43	9.7	0	0	0	550	50	3.5	40	No	Horizontal Tube	59.21	12
43	43	9.7	0	0	0	550	20	3.5	40	No	Horizontal Tube	59.7	12
43	43	9.7	0	0	0	450	50	3.5	40	No	Horizontal Tube	48.38	12
43	43	9.7	0	0	0	550	20	3.5	40	No	Horizontal Tube	60.48	12
43	43	9.7	0	0	0	450	50	3.5	40	No	Horizontal Tube	13.61	12
43	43	9.7	0	0	0	450	20	3.5	40	No	Horizontal Tube	46.79	12
43	43	9.7	0	0	0	450	20	3.5	40	No	Horizontal Tube	15.4	12
43	43	9.7	0	0	0	550	35	3.5	40	No	Horizontal Tube	58.97	12
43	43	9.7	0	0	0	450	35	3.5	40	No	Horizontal Tube	34.94	12
43	43	9.7	0	0	0	500	50	3.5	40	No	Horizontal Tube	63.41	12
43	43	9.7	0	0	0	500	20	3.5	40	No	Horizontal Tube	66.51	12
43	43	9.7	0	0	0	500	35	3.5	40	No	Horizontal Tube	64.56	12
43	43	9.7	0	0	0	500	35	3.5	40	No	Horizontal Tube	64.22	12
29.3	29.3	26.9	8.8	0	5.6	500	10	-	-	No	Batch	75.8	13
29.3	29.3	26.9	8.8	0	5.6	500	20	-	-	No	Batch	82	13
0	0	100	0	0	0	250	-	-	2	No	Batch	57.27	14
0	0	100	0	0	0	300	-	-	2	No	Batch	69.82	14
0	0	100	0	0	0	350	-	-	2	No	Batch	67.74	14
0	0	100	0	0	0	400	-	-	2	No	Batch	63.23	14
100	0	0	0	0	0	300	-	-	2	No	Batch	30.7	14
100	0	0	0	0	0	350	-	-	2	No	Batch	80.88	14
100	0	0	0	0	0	400	-	-	2	No	Batch	54.17	14
0	0	100	0	0	0	460	-	-	10	No	Batch	86	15
0	34	66	0	0	0	460	-	-	10	No	Batch	84	15
0	66	34	0	0	0	460	-	-	10	No	Batch	63	15

HDPE (wt%)	LDPE (wt%)	PP (wt%)	PS (wt%)	PVC (wt%)	PET (wt%)	Temperature (°C)	Heating Rate (°C/min)	Particle Size (mm)	Feed Size (g)	Catalyst	Reactor Type	Oil Yield (wt%)	Source
0	100	0	0	0	0	460	-	-	10	No	Batch	95	15
0	66	34	0	0	0	460	10	-	10	Yes	Batch	67	15
0	34	66	0	0	0	460	10	-	10	Yes	Batch	62	15
0	0	100	0	0	0	460	10	-	10	Yes	Batch	57	15
0	100	0	0	0	0	460	10	-	10	Yes	Batch	50	15
0	100	0	0	0	0	500	6	2	-	No	Batch	80.41	16
0	100	0	0	0	0	500	8	2	-	No	Batch	79.64	16
0	100	0	0	0	0	500	10	2	-	No	Batch	76.45	16
0	100	0	0	0	0	500	12	2	-	No	Batch	74.32	16
0	100	0	0	0	0	500	14	2	-	No	Batch	71.11	16
0	0	0	0	0	100	500	6	2	-	No	Batch	38.89	16
0	0	0	0	0	100	500	8	2	-	No	Batch	34.16	16
0	0	0	0	0	100	500	10	2	-	No	Batch	32.13	16
0	0	0	0	0	100	500	12	2	-	No	Batch	30.33	16
0	0	0	0	0	100	500	14	2	-	No	Batch	29.14	16
0	0	100	0	0	0	500	6	2	-	No	Batch	82.12	16
0	0	100	0	0	0	500	8	2	-	No	Batch	81.32	16
0	0	100	0	0	0	500	10	2	-	No	Batch	80.65	16
0	0	100	0	0	0	500	12	2	-	No	Batch	79.41	16
0	0	100	0	0	0	500	14	2	-	No	Batch	78.26	16
100	0	0	0	0	0	430	3	-	10	No	Batch	84.7	17
0	100	0	0	0	0	430	3	-	10	No	Batch	84.3	17
0	100	0	0	0	0	430	3	-	10	No	Batch	84.6	17
50	50	0	0	0	0	430	3	-	10	No	Batch	83.7	17
0	0	0	100	0	0	450	10	-	1000	No	Batch	80.8	18
50	50	0	0	0	0	450	10	-	1000	No	Batch	25	18

HDPE (wt%)	LDPE (wt%)	PP (wt%)	PS (wt%)	PVC (wt%)	PET (wt%)	Temperature (°C)	Heating Rate (°C/min)	Particle Size (mm)	Feed Size (g)	Catalyst	Reactor Type	Oil Yield (wt%)	Source
0	0	100	0	0	0	450	10	-	1000	No	Batch	42	18
0	0	50	50	0	0	450	10	-	1000	No	Batch	25	18
25	25	0	50	0	0	450	10	-	1000	No	Batch	54	18
25	25	50	0	0	0	450	10	-	1000	No	Batch	24	18
12.5	12.5	25	50	0	0	450	10	-	1000	No	Batch	49	18
10	10	20	40	0	20	450	10	-	1000	No	Batch	40	18
0	100	0	0	0	0	450	7.5	-	17	No	Batch	79.8	19
100	0	0	0	0	0	450	7.5	-	17	No	Batch	79.8	19
0	100	0	0	0	0	450	7.5	-	17	No	Batch	79.1	19
0	0	100	0	0	0	450	7.5	-	17	No	Batch	85	19
0	50	50	0	0	0	450	7.5	-	17	No	Batch	79.4	19
33.3	33.3	33.3	0	0	0	450	7.5	-	17	No	Batch	82.4	19
25	50	25	0	0	0	450	7.5	-	17	No	Batch	79.7	19
34.6	34.6	9.6	9.6	10.6	1.1	450	7.5	-	17	No	Batch	75.4	19
25	50	25	0	0	0	450	-	-	17	No	Batch	79.7	20
0	100	0	0	0	0	450	-	-	17	No	Batch	79.1	20
44.4	0	21.2	13.3	12.2	8.9	500	5	7.5	35	No	Batch	48.7	21
100	0	0	0	0	0	500	5	3.5	35	No	Batch	93	21
0	0	100	0	0	0	500	5	3.5	35	No	Batch	95	21
0	0	0	100	0	0	500	5	3.5	35	No	Batch	71	21
0	0	0	0	0	100	500	5	3.5	35	No	Batch	15	21
0	100	0	0	0	0	500	5	3	-	No	Fluidized Bed	89.2	22
0	100	0	0	0	0	550	5	3	-	No	Fluidized Bed	78.6	22
0	100	0	0	0	0	600	5	3	-	No	Fluidized Bed	75.8	22
0	100	0	0	0	0	650	5	3	-	No	Fluidized Bed	59.9	22
0	100	0	0	0	0	700	5	3	-	No	Fluidized Bed	28.6	22

HDPE (wt%)	LDPE (wt%)	PP (wt%)	PS (wt%)	PVC (wt%)	PET (wt%)	Temperature (°C)	Heating Rate (°C/min)	Particle Size (mm)	Feed Size (g)	Catalyst	Reactor Type	Oil Yield (wt%)	Source
100	0	0	0	0	0	640	10	-	-	No	Fluidized Bed	68.5	23
100	0	0	0	0	0	680	10	-	-	No	Fluidized Bed	39.6	23
100	0	0	0	0	0	730	10	-	-	No	Fluidized Bed	18	23
100	0	0	0	0	0	780	10	-	-	No	Fluidized Bed	9.6	23
100	0	0	0	0	0	850	10	-	-	No	Fluidized Bed	16.2	23
50	50	0	0	0	0	500	10	5	2	No	Fixed Bed	83	24
0	0	100	0	0	0	500	10	5	2	No	Fixed Bed	81	24
0	0	0	100	0	0	500	10	5	2	No	Fixed Bed	97	24
0	0	0	0	0	100	500	10	5	2	No	Fixed Bed	38.5	24
30	30	13	18	0	0	500	10	5	2	No	Fixed Bed	80	24
30	30	13	18	0	0	500	10	5	2	No	Fixed Bed	88	24
100	0	0	0	0	0	500	10	-	2	Yes	Fixed Bed	57	24
0	0	100	0	0	0	500	10	-	2	Yes	Fixed Bed	51	24
0	0	0	100	0	0	500	10	-	2	Yes	Fixed Bed	93	24
0	0	0	0	0	100	500	10	-	2	Yes	Fixed Bed	43	24
30	30	13	18	0	0	500	10	-	2	Yes	Fixed Bed	51	24
30	30	13	18	0	0	500	10	-	2	Yes	Fixed Bed	57	24
29.4	29.4	26.9	8.7	0	5.6	450	20	20	200	No	Batch	78	25
29.4	29.4	26.9	8.7	0	5.6	500	20	20	200	No	Batch	80.5	25
29.4	29.4	26.9	8.7	0	5.6	550	20	20	200	No	Batch	82	25
29.4	29.4	26.9	8.7	0	5.6	600	20	20	200	No	Batch	84	25
100	0	0	0	0	0	645	-	0.225	-	No	Fluidized Bed	79.7	26
100	0	0	0	0	0	640	-	0.225	-	No	Fluidized Bed	78.9	26
100	0	0	0	0	0	650	-	0.225	-	No	Fluidized Bed	68.5	26
100	0	0	0	0	0	650	-	0.225	-	No	Fluidized Bed	72.3	26
100	0	0	0	0	0	685	-	0.225	-	No	Fluidized Bed	33.4	26

HDPE (wt%)	LDPE (wt%)	PP (wt%)	PS (wt%)	PVC (wt%)	PET (wt%)	Temperature (°C)	Heating Rate (°C/min)	Particle Size (mm)	Feed Size (g)	Catalyst	Reactor Type	Oil Yield (wt%)	Source
100	0	0	0	0	0	685	-	0.225	-	No	Fluidized Bed	39.6	26
100	0	0	0	0	0	700	-	0.225	-	No	Fluidized Bed	32.1	26
100	0	0	0	0	0	685	-	0.225	-	No	Fluidized Bed	40.7	26
100	0	0	0	0	0	730	-	0.225	-	No	Fluidized Bed	19.6	26
100	0	0	0	0	0	725	-	0.225	-	No	Fluidized Bed	20.7	26
100	0	0	0	0	0	715	-	0.225	-	No	Fluidized Bed	19.2	26
100	0	0	0	0	0	730	-	0.225	-	No	Fluidized Bed	13.5	26
100	0	0	0	0	0	780	-	0.225	-	No	Fluidized Bed	15.3	26
100	0	0	0	0	0	780	-	0.225	-	No	Fluidized Bed	9.6	26
100	0	0	0	0	0	780	-	0.225	-	No	Fluidized Bed	13.4	26
100	0	0	0	0	0	800	-	0.225	-	No	Fluidized Bed	13.7	26
100	0	0	0	0	0	850	-	0.225	-	No	Fluidized Bed	11.4	26
100	0	0	0	0	0	850	-	0.225	-	No	Fluidized Bed	16.2	26
100	0	0	0	0	0	850	-	0.225	-	No	Fluidized Bed	15.4	26
100	0	0	0	0	0	850	-	0.225	-	No	Fluidized Bed	12.2	26
0	0	0	100	0	0	450	5	3.5	10	No	Fluidized Bed	97.6	27
0	0	0	100	0	0	500	5	3.5	10	No	Fluidized Bed	96.4	27
0	0	0	100	0	0	550	5	3.5	10	No	Fluidized Bed	95.3	27
0	0	0	100	0	0	600	5	3.5	10	No	Fluidized Bed	98.7	27
0	0	0	100	0	0	650	5	3.5	10	No	Fluidized Bed	90.7	27
0	0	0	100	0	0	700	5	3.5	10	No	Fluidized Bed	90.2	27
0	100	0	0	0	0	425	10	2	10	No	Batch	89.5	28
0	100	0	0	0	0	450	10	2	10	No	Batch	72.4	28
0	100	0	0	0	0	500	10	2	10	No	Batch	37.5	28
0	0	0	100	0	0	350	10	2	10	No	Batch	99	28
0	0	0	100	0	0	450	10	2	10	No	Batch	79.4	28

HDPE (wt%)	LDPE (wt%)	PP (wt%)	PS (wt%)	PVC (wt%)	PET (wt%)	Temperature (°C)	Heating Rate (°C/min)	Particle Size (mm)	Feed Size (g)	Catalyst	Reactor Type	Oil Yield (wt%)	Source
0	0	0	100	0	0	500	10	2	10	No	Batch	67.1	28
0	70	0	30	0	0	400	10	2	10	No	Batch	96	28
0	70	0	30	0	0	425	10	2	10	No	Batch	90.2	28
0	70	0	30	0	0	450	10	2	10	No	Batch	83.7	28
0	0	100	0	0	0	380	3	-	10	No	Batch	64.9	29
0	0	100	0	0	0	380	3	-	10	No	Batch	80.1	29
0	0	100	0	0	0	380	3	-	10	Yes	Batch	54.5	29
0	0	100	0	0	0	380	3	-	10	Yes	Batch	68.8	29
0	0	100	0	0	0	380	3	-	10	Yes	Batch	78.3	29
0	0	100	0	0	0	380	3	-	10	Yes	Batch	47	29
0	0	100	0	0	0	380	3	-	10	Yes	Batch	75.4	29
0	0	100	0	0	0	380	3	-	10	Yes	Batch	80.6	29
0	0	100	0	0	0	380	3	-	10	Yes	Batch	86.4	29
0	0	100	0	0	0	380	3	-	10	Yes	Batch	85.4	29
0	0	100	0	0	0	380	3	-	10	Yes	Batch	83.3	29
0	0	100	0	0	0	380	3	-	10	Yes	Batch	83.7	29
0	0	100	0	0	0	380	3	-	10	Yes	Batch	83	29
100	0	0	0	0	0	430	3	-	10	No	Batch	69.3	29
100	0	0	0	0	0	430	3	-	10	Yes	Batch	67.8	29
100	0	0	0	0	0	430	3	-	10	Yes	Batch	74.3	29
100	0	0	0	0	0	430	3	-	10	Yes	Batch	49.8	29
100	0	0	0	0	0	430	3	-	10	Yes	Batch	71.1	29
100	0	0	0	0	0	430	3	-	10	Yes	Batch	81.4	29
100	0	0	0	0	0	430	3	-	10	Yes	Batch	80.1	29
100	0	0	0	0	0	430	3	-	10	Yes	Batch	81.9	29
100	0	0	0	0	0	430	3	-	10	Yes	Batch	75.2	29

HDPE (wt%)	LDPE (wt%)	PP (wt%)	PS (wt%)	PVC (wt%)	PET (wt%)	Temperature (°C)	Heating Rate (°C/min)	Particle Size (mm)	Feed Size (g)	Catalyst	Reactor Type	Oil Yield (wt%)	Source
0	0	100	0	0	0	250	-	50	1000	No	Batch	51.76	30
0	0	100	0	0	0	300	-	50	1000	No	Batch	68.98	30
0	0	100	0	0	0	350	-	50	1000	No	Batch	79.68	30
0	0	100	0	0	0	400	-	50	1000	No	Batch	80.14	30
0	0	100	0	0	0	250	-	50	1000	No	Batch	52.31	30
0	0	100	0	0	0	300	-	50	1000	No	Batch	69.32	30
0	0	100	0	0	0	350	-	50	1000	No	Batch	85.06	30
0	0	100	0	0	0	400	-	50	1000	No	Batch	88.86	30
25	35	40	0	0	0	400	12	1.4	-	No	Fixed Bed	18.89	31
25	35	40	0	0	0	500	12	1.4	-	No	Fixed Bed	30.66	31
25	35	40	0	0	0	450	12	1.4	-	No	Fixed Bed	26.68	31
25	35	40	0	0	0	475	12	1.4	-	No	Fixed Bed	28.26	31
25	35	40	0	0	0	500	12	1.4	-	No	Fixed Bed	32.8	31
25	35	40	0	0	0	525	12	1.4	-	No	Fixed Bed	28.8	31
25	35	40	0	0	0	500	12	1.4	-	No	Fixed Bed	28.8	31
25	35	40	0	0	0	500	12	1.4	-	No	Fixed Bed	30.37	31
40	0	35	18	4	3	460	20	3	100	No	Semi Batch	72	32
40	0	35	18	4	3	500	20	3	100	No	Semi Batch	65.2	32
40	0	35	18	4	3	600	20	3	100	No	Semi Batch	42.9	32
0	100	0	0	0	0	300	-	3.5	100	Yes	Fixed Bed	53.5	33
0	90	10	0	0	0	560	10	4	22.5	Yes	Batch	29	34
0	70	30	0	0	0	560	10	4	22.5	Yes	Batch	68	34
0	50	50	0	0	0	560	10	4	22.5	Yes	Batch	70	34
0	30	70	0	0	0	560	10	4	22.5	Yes	Batch	78	34
0	10	90	0	0	0	560	10	4	22.5	Yes	Batch	80	34
10	90	0	0	0	0	560	10	4	22.5	Yes	Batch	55	34

HDPE (wt%)	LDPE (wt%)	PP (wt%)	PS (wt%)	PVC (wt%)	PET (wt%)	Temperature (°C)	Heating Rate (°C/min)	Particle Size (mm)	Feed Size (g)	Catalyst	Reactor Type	Oil Yield (wt%)	Source
30	70	0	0	0	0	560	10	4	22.5	Yes	Batch	75	34
50	50	0	0	0	0	560	10	4	22.5	Yes	Batch	78	34
70	30	0	0	0	0	560	10	4	22.5	Yes	Batch	75	34
90	10	0	0	0	0	560	10	4	22.5	Yes	Batch	61	34
10	0	0	90	0	0	560	10	4	22.5	Yes	Batch	81	34
30	0	0	70	0	0	560	10	4	22.5	Yes	Batch	68	35
50	0	0	50	0	0	560	10	4	22.5	Yes	Batch	70	35
70	0	0	30	0	0	560	10	4	22.5	Yes	Batch	80	35
90	0	0	10	0	0	560	10	4	22.5	Yes	Batch	81	35
0	0	100	0	0	0	500	20	2.5	20	No	Batch	83	35
0	0	100	0	0	0	500	20	2.5	20	Yes	Batch	84	35
0	0	100	0	0	0	500	20	2.5	20	Yes	Batch	85	35
0	0	100	0	0	0	500	20	2.5	20	Yes	Batch	83	35
0	0	100	0	0	0	500	20	2.5	20	Yes	Batch	77	35
0	100	0	0	0	0	500	20	2.5	20	No	Batch	76	35
0	100	0	0	0	0	500	20	2.5	20	Yes	Batch	78	35
0	100	0	0	0	0	500	20	2.5	20	Yes	Batch	82	35
0	100	0	0	0	0	500	20	2.5	20	Yes	Batch	75	35
0	100	0	0	0	0	500	20	2.5	20	Yes	Batch	68	35
100	0	0	0	0	0	500	20	2.5	20	No	Batch	78	35
100	0	0	0	0	0	500	20	2.5	20	Yes	Batch	79	35
100	0	0	0	0	0	500	20	2.5	20	Yes	Batch	80	35
100	0	0	0	0	0	500	20	2.5	20	Yes	Batch	79	35
100	0	0	0	0	0	500	20	2.5	20	Yes	Batch	73	35
33.3	33.3	33.3	0	0	0	500	20	2.5	20	No	Batch	77	35
33.3	33.3	33.3	0	0	0	500	20	2.5	20	Yes	Batch	79	35

HDPE (wt%)	LDPE (wt%)	PP (wt%)	PS (wt%)	PVC (wt%)	PET (wt%)	Temperature (°C)	Heating Rate (°C/min)	Particle Size (mm)	Feed Size (g)	Catalyst	Reactor Type	Oil Yield (wt%)	Source
33.3	33.3	33.3	0	0	0	500	20	2.5	20	Yes	Batch	80	35
33.3	33.3	33.3	0	0	0	500	20	2.5	20	Yes	Batch	78	35
33.3	33.3	33.3	0	0	0	500	20	2.5	20	Yes	Batch	76	35
0	0	100	0	0	0	400	-	25	1000	No	Batch	78	36
0	0	100	0	0	0	450	-	25	1000	No	Batch	79.6	36
0	0	100	0	0	0	500	-	25	1000	No	Batch	81	36
0	0	100	0	0	0	400	-	25	1000	Yes	Batch	78.4	36
0	0	100	0	0	0	450	-	25	1000	Yes	Batch	75.5	36
0	0	100	0	0	0	500	-	25	1000	Yes	Batch	53.2	36
0	0	90	0	0	10	400	-	25	1000	Yes	Batch	73	36
0	0	75	0	0	25	450	-	25	1000	Yes	Batch	54.7	36
0	0	65	0	0	35	500	-	25	1000	Yes	Batch	45	36
0	0	0	100	0	0	450	5	20	1000	Yes	Batch	70	37
0	0	0	100	0	0	450	10	20	1000	Yes	Batch	60	37
50	50	0	0	0	0	450	10	20	1000	Yes	Batch	40	37
50	50	0	0	0	0	450	10	20	1000	Yes	Batch	42	37
0	0	100	0	0	0	450	10	20	1000	Yes	Batch	40	37
0	0	100	0	0	0	450	10	20	1000	Yes	Batch	54	37
0	50	0	50	0	0	450	10	20	1000	Yes	Batch	44	37
0	50	0	50	0	0	450	10	20	1000	Yes	Batch	52	37
0	0	50	50	0	0	450	10	20	1000	Yes	Batch	54	37
0	0	50	50	0	0	450	10	20	1000	Yes	Batch	34	37
0	50	50	0	0	0	450	10	20	1000	Yes	Batch	44	37
0	50	50	0	0	0	450	10	20	1000	Yes	Batch	40	37
0	25	25	50	0	0	450	10	20	1000	Yes	Batch	44	37
0	25	25	50	0	0	450	10	20	1000	Yes	Batch	40	37

HDPE (wt%)	LDPE (wt%)	PP (wt%)	PS (wt%)	PVC (wt%)	PET (wt%)	Temperature (°C)	Heating Rate (°C/min)	Particle Size (mm)	Feed Size (g)	Catalyst	Reactor Type	Oil Yield (wt%)	Source
0	20	20	20	0	20	450	10	20	1000	Yes	Batch	28	37
0	20	20	20	0	20	450	10	20	1000	Yes	Batch	30	37
0	100	0	0	0	0	450	55	0.5	15	Yes	Semi Batch	34.5	38
0	100	0	0	0	0	500	55	0.5	15	Yes	Semi Batch	56.53	38
0	100	0	0	0	0	550	55	0.5	15	Yes	Semi Batch	53	38
0	100	0	0	0	0	600	55	0.5	15	Yes	Semi Batch	48.97	38
0	17	0	17	0	66	450	6	100	60	No	Semi Batch	10	39
0	17	0	66	0	17	450	6	100	60	No	Semi Batch	36	39
0	100	0	0	0	0	450	6	100	60	No	Semi Batch	30	39
0	100	0	0	0	0	450	6	100	60	No	Semi Batch	32	39
0	0	0	0	0	100	450	6	100	60	No	Semi Batch	4	39
0	50	0	0	0	50	450	6	100	60	No	Semi Batch	8	39
0	33	0	33	0	33	450	6	100	60	No	Semi Batch	33	39
0	50	0	50	0	0	450	6	100	60	No	Semi Batch	52	39
0	0	0	0	0	100	450	6	100	60	No	Semi Batch	2	39
0	0	0	100	0	0	450	6	100	60	No	Semi Batch	49	39
0	0	0	50	0	50	450	6	100	60	No	Semi Batch	27	39
0	67	0	17	0	17	450	6	100	60	No	Semi Batch	34	39
0	0	0	100	0	0	450	6	100	60	No	Semi Batch	40	39

Section SI.2 Initial Testing

Preliminary tests eliminated all data that had a data gap in one or more features, resulting in a dataset consisting of 149 data points. Although the corresponding MAE of the stripped-down validation set was 8.0% for Random Forest and 8.5% for XGBoost (less than what was found for the final models recommended here), the standard deviations on the validation sets were 1.5% and $\pm 1.8\%$, much greater than the recommended models which indicates a strong dependence on the data used for training. Further, the test set MAE values were 9.9% for RF and 11.5% for XGBoost, and the corresponding differences between validation and test set MAE were 1.9 and 3, much greater than observed for the final models recommended here. Taken together, these observations indicate that the stripped-down data set led to a much stronger reliance on selection of training data and a much greater propensity for over fitting than observed in the models recommended here.

Instead of data removal and to develop more robust models that were less prone to overfitting, the K nearest neighbor (KNN) method was used to populate gaps in the reported values of features.

Further testing that tried to conflate the different reactor types or reaction times yielded a Random Forest validation set MAE of 9.6% and a test set MAE of 8.8%. The fact that the test set MAE is lower than the validation set indicates in this case that the model has been over specified and the accuracy of the predictions cannot be trusted.

Section SI.3 Model Parameters

Each of the seven models studied in this work were optimized such that their hyper parameters gave rise to the minimum MAE for that model. The optimal hyper parameters can be seen below for each of the seven models in Table SI.2

Table SI.2 Optimal Model Hyper Parameters for Seven Models Studied

<i>Model</i>	<i>Hyper Parameters</i>			
<i>Linear Regression</i>	Random State			
	50			
<i>Lasso Regression</i>	Random State Alpha Max iterations			
	10 0.0001 5000			
<i>Ridge Regression</i>	Random State Alpha Max iterations			
	50 0.001 5000			
<i>Decision Tree</i>	Random State Max Features Max Depth			
	55 11 16			
<i>XGBoost</i>	Random State Number of Estimators Learning Rate			
	50 640 0.05			
<i>Random Forest</i>	Random State Number of Estimators Max Features Max Depth			
	100 108 5 16			
<i>Artificial Neural Network</i>	Random State Max Iterations Hidden Layer Sizes Batch Size			
	25 3000 (640,) 190			

Section SI.4 Additional Model Results and Figures

Figure SI.1 shows the parity plot for the reported experimental oil yield vs the oil yields predicted by the optimal XGBoost model, along with the 1:1 line and the $\pm 10\%$ region.

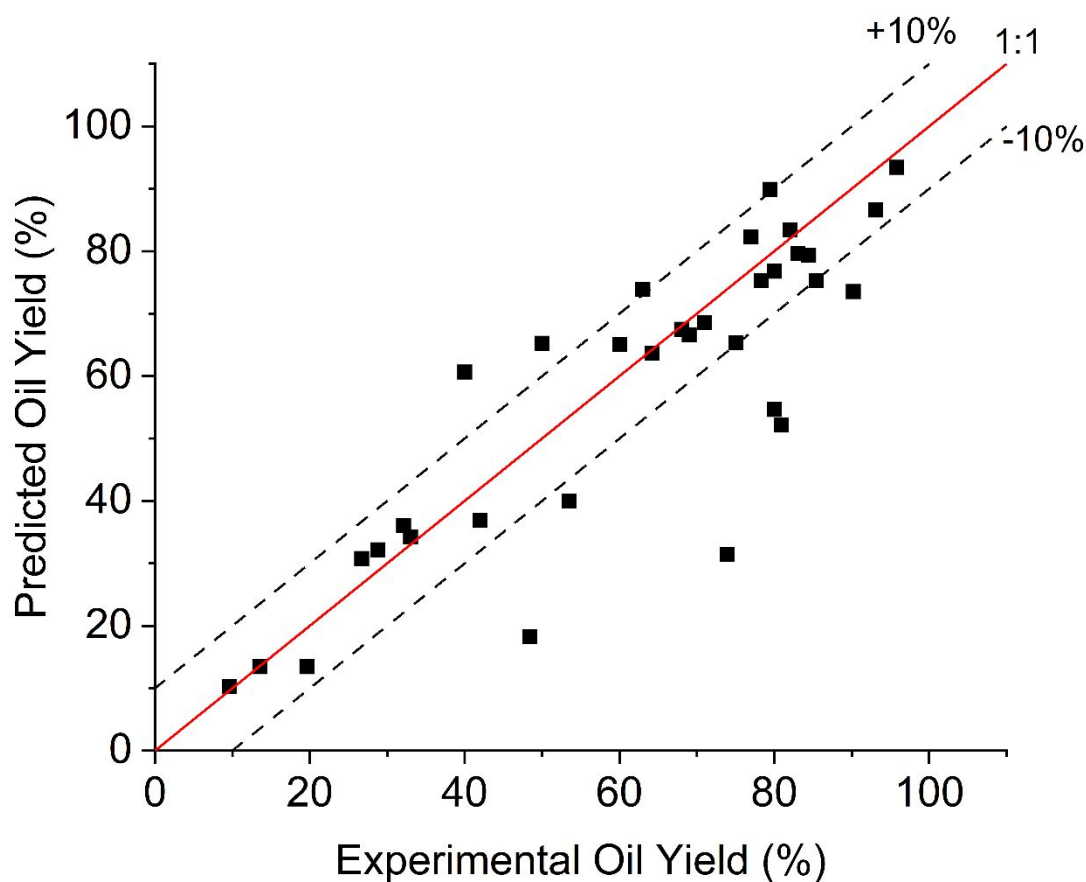


Figure SI.1 Parity plot for the reported experimental oil yield vs the oil yield predicted by the optimal XGBoost model for the test set.

For the data points in the test set that lay outside of the $\pm 10\%$ region Table SI.3 evaluates the reaction conditions in order to determine if there was any systematic error. As can be seen from this table there is no trend that can be identified and no indication of any systematic error.

Table SI.3 Analysis of conditions for outliers from Figure SI.1

<i>Difference</i>	<i>Plastic Type</i>	<i>Temperature</i>	<i>Reactor</i>	<i>Catalyst</i>	<i># of KNN</i>
<i>in</i>			<i>Type</i>		<i>Predicted</i>
<i>Prediction</i>					<i>Features</i>

16.6	PS	700	Fluidized Bed	No	0
28.7	HDPE	350	Batch	No	2
25.4	LDPE	500	Fixed Bed	Yes	1
-10.5	PS	450	Batch	No	0
42.4	LDPE	550	Fluidized Bed	No	2
-10.9	2:1 LDPE:PP	460	Batch	No	2
13.5	LDPE	300	Fixed Bed	Yes	1
-15.3	LDPE	460	Batch	Yes	1
30.1	43:43:9.7 HDPE:LDPE:PP	450	Horizontal Tube	No	0
-20.6	PP	450	Batch	Yes	0

Figure SI. 2 shows the feature importance of each of the model features from the optimal XGBoost model. The plastic types each have their own feature importance but were grouped in Figure SI.2 due to their relatively small values and for simplicity.

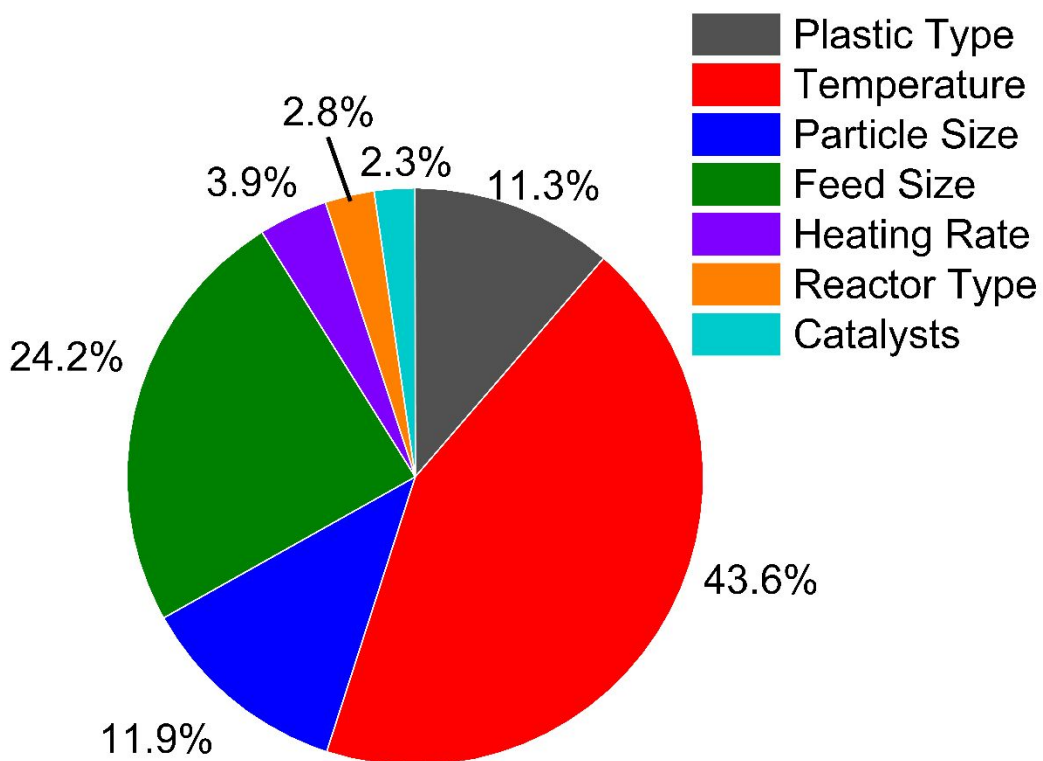


Figure SI.2 Feature Importance for each feature included in the final optimal XGBoost Model. Plastic types are lumped into one category for simplicity

Figure SI.3 shows the oil yields predicted for pyrolysis of pure plastics at 500 °C. The other features were chosen as representative values and are found in Table SI.4.

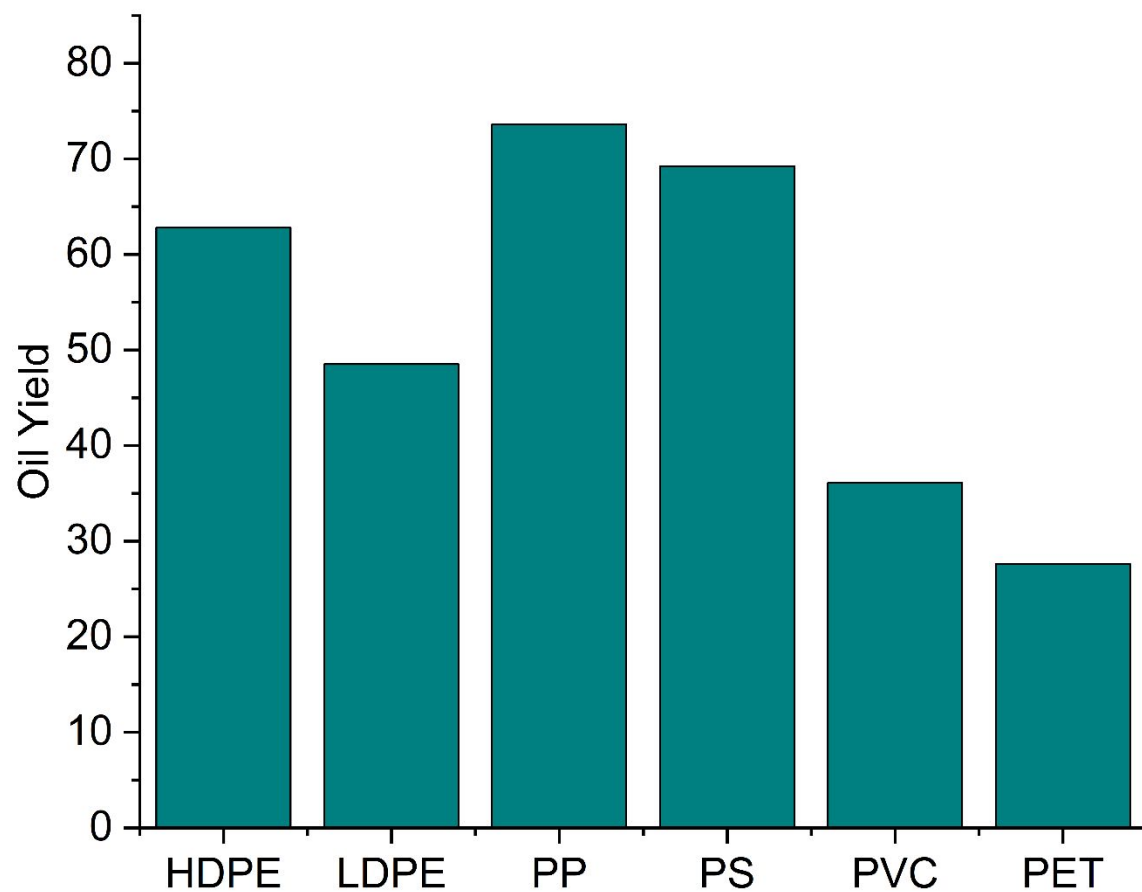


Figure SI.3 Predict oil yield (mass percent) for the pyrolysis of pure plastics at 500 °C

Figure SI.4 shows the effect of PET on a mixture of PP and LDPE.

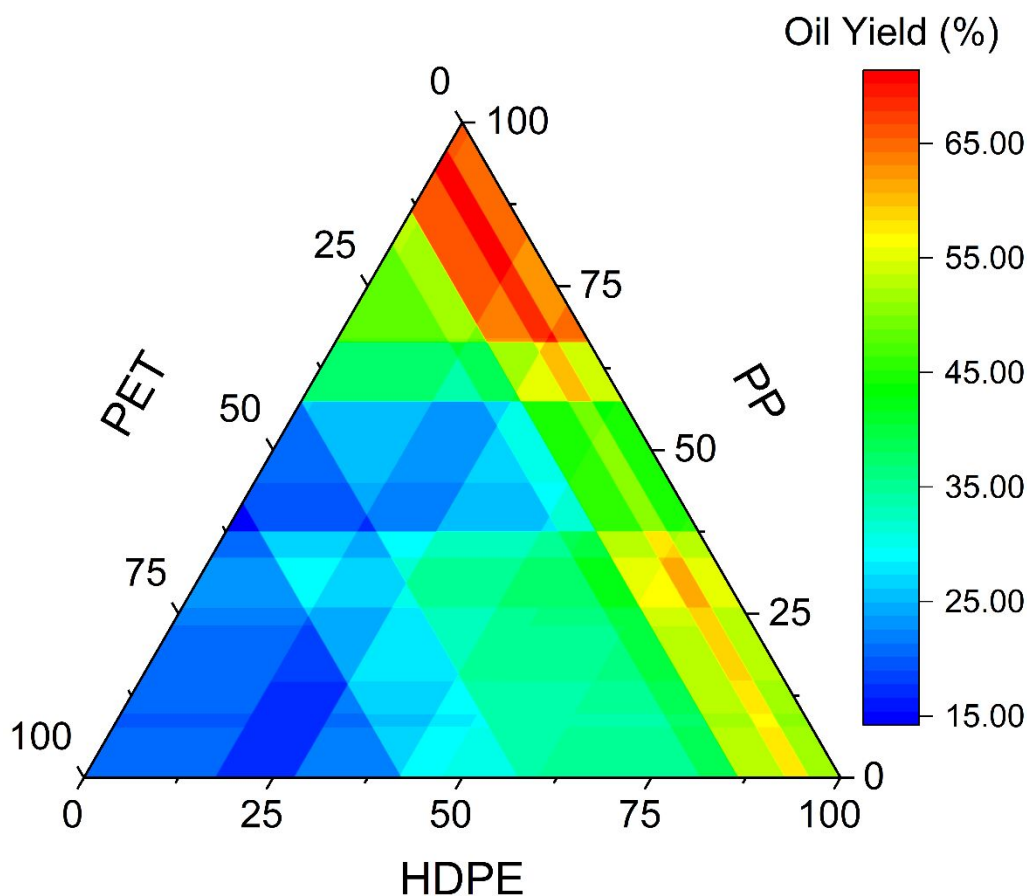


Figure SI.4 Predicted oil yields for mixtures of LDPE, PP and PET pyrolyzed in a horizontal tube reactor at 500 °C, with a heating rate of 10 °C/min, a particle size of 13mm, plastics loading of 200g without catalyst.

Table SI.4 and Table SI.5 provide the feed composition (Table SI.3) and reaction parameters (Table IS.4) used for the prediction of oil yields from the pyrolysis of real waste plastic found in the Rhine River in Europe.

Table SI.4 Composition of Plastic Found in the Rhine River ^{40, 41}

<i>Plastic Type</i>	<i>Weight Percent in Mixture (%)</i>
<i>HDPE</i>	27
<i>LDPE</i>	0
<i>PP</i>	37
<i>PS</i>	26
<i>PVC</i>	2
<i>PET</i>	0

Table SI.5 Estimated Reaction Parameters Used for Rhine River Oil Yields

<i>Feature</i>	<i>Value</i>
<i>Particle Size</i>	13 mm
<i>Heating Rate</i>	10 °C/min
<i>Feed Size</i>	200 g
<i>Catalyst</i>	None
<i>Reactor Type</i>	Horizontal Tube

Section SI.5 Thermodynamic Model Modifications

The Monte Carlo (MC) model previously developed by the group ⁴² utilized oil yields and specific oil composition from data published in the literature. In this work the specific composition of the oil products is not known as new feeds are studied. In order to handle this, the exergy of combustion of the oil is calculated using an estimated HHV of the oil instead of the specific heats of combustion of the products. An average HHV of 41 MJ/kg was used and varied by $\pm 10\%$ to capture how the feed composition could affect the HHV of the oil ⁴³. Other necessary modifications to the model included removing one of the water filter steps and removal of the RO as these were not necessary in fresh water. Finally, as the system would be stationary and not operating on a ship the exergy required for the boat engine was removed from the model. The full list of stochastic variables used in the MC simulation can be found in Table SI.6 and the constant variables used in the model can be found in Table SI.7.

Table SI.6 Uniform distribution of stochastic variables used in the MC model

<i>Variable</i>	<i>Minimum Value</i>	<i>Maximum Value</i>
<i>Oil Yield (%) at 400 °C</i>	47.6	65.8
<i>Oil Yield (%) at 450 °C</i>	43.9	62.1
<i>Oil Yield (%) at 500 °C</i>	46.1	64.3

<i>Oil Yield (%) at 550 °C</i>	36.5	54.7
<i>Oil Yield (%) at 600 °C</i>	35.6	53.8
<i>Oil Yield (%) at 650 °C</i>	35	53.2
<i>HHV (MJ/kg)</i>	36.9	45.1
<i>Heat Exchanger efficiency</i>	0.5	0.8
<i>Dryer Efficiency</i>	0.4	0.6
<i>Ambient Water Temperature (K)</i>	291	296
<i>Heat of Combustion Variance</i>	0.98	1.02

Table SI.7 Constant Variables Energy Draw in the MC Model

<i>Variable</i>	<i>Value</i>
<i>Pump (kW)</i>	0.7457
<i>Shredder (kW)</i>	2.2
<i>Conveyor Belt (kW)</i>	0.372
<i>Exergy Difference b/w polymer and fuel (kW)</i>	1.36

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